

Problem Statement #15, Pharmacy Expiry & Re-order Dispatch Engine

NAME: Siri Krishna C R

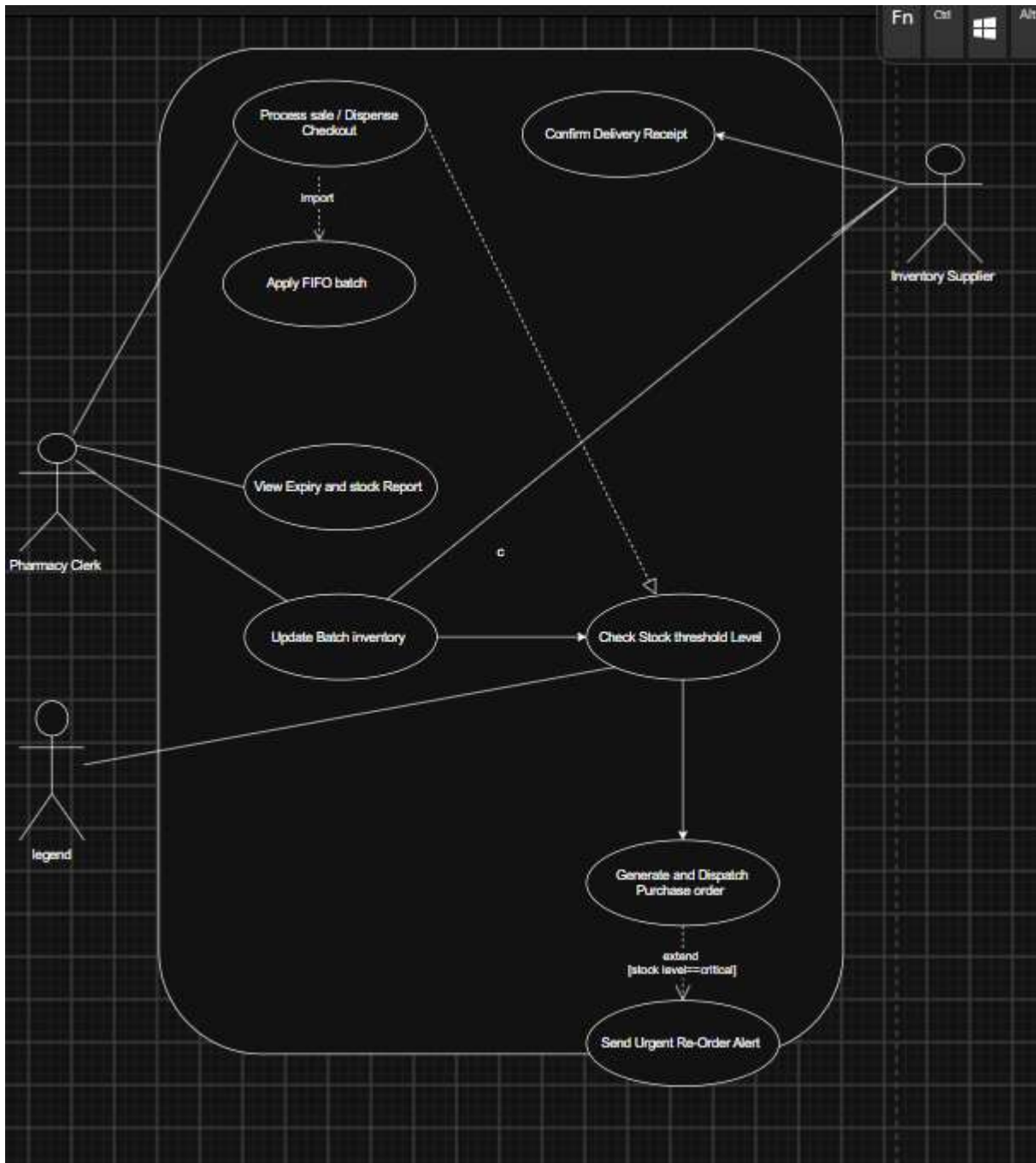
SRN: PES1UG24AM276

Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce First-Expired-First-Out (FEFO) picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	High	Pass: Oldest valid (non-expired) batch is auto-selected first. Fail: A newer batch is assigned while an older, unexpired batch of the same SKU still has stock.	Minimizes medicine wastage from expiry and reduces the risk of dispensing near-expiry stock to patients.
FR-002	Functional	The system shall continuously monitor per-SKU stock levels against a configurable safety threshold and flag any SKU that falls below it.	High	Pass: SKU is flagged within 5 minutes of stock dropping below threshold. Fail: SKU remains unflagged after threshold breach or is flagged incorrectly while above threshold.	Enables proactive replenishment and prevents stock-outs of critical medicines.
FR-003	Functional	The system shall automatically generate and dispatch a purchase order to the relevant Inventory Supplier when a flagged SKU's stock dips below its safety threshold.	High	Pass: A correctly itemized PO (SKU, quantity, supplier) is dispatched with no clerk intervention. Fail: PO is missing, duplicated, or sent to the wrong supplier.	Removes manual re-order effort and shortens replenishment lead time.
FR-004	Functional	The system shall flag batches nearing expiry within a configurable N-day window (default 30 days) and surface them in a near-expiry report visible to the Pharmacy Clerk.	Medium	Pass: All batches expiring within the configured window appear on the report the same day they cross the window. Fail: A near-expiry batch is absent from the report.	Gives clerks early visibility to prioritize dispensing or return/discard slow-moving stock before it expires.
FR-005	Functional	The system shall allow the Pharmacy Clerk or Inventory Supplier to confirm receipt of a dispatched purchase order, upon which the corresponding batch (with quantity and expiry date) is added to inventory.	Medium	Pass: Confirming a delivery updates stock count and creates a new batch record with the correct expiry date. Fail: Inventory is not updated, or is updated with the wrong batch/expiry.	Keeps stock records accurate and closes the loop between ordering and replenishment.
NFR-001	Performance & Security	The system shall generate and dispatch supplier purchase order requests within 2 seconds of a threshold breach being detected, over an encrypted (TLS 1.2+) channel with supplier authentication.	High	Pass: Benchmarking under simulated peak load (500 concurrent SKU checks) confirms PO dispatch latency under 2 seconds and all traffic is encrypted with verified supplier credentials. Fail: Latency exceeds target or any PO is sent over an unencrypted/unauthenticated channel.	Slow or insecure re-order dispatch risks stock-outs of critical medicines and exposes sensitive supplier/pricing data.
NFR-002	Reliability & Data Integrity	The system shall maintain 99.9% monthly uptime for checkout and FEFO-selection operations, and shall	High	Pass: Uptime monitoring logs confirm 99.9% availability over a 30-day window; a simulated node failure during a checkout write results in zero data loss on failover. Fail: Uptime falls	Pharmacy checkout is a continuous, patient-facing operation; downtime or data loss can halt

ID	Type	Description	Priority Acceptance Criteria	Rationale
		guarantee zero loss of batch/expiry data during any single-node failure through redundant storage.	below target, or any batch/expiry record is lost after a simulated failure.	dispensing or cause incorrect FEFO decisions.

UML Use-Case Diagram



Use-Case Flow Specification: Process Sale / Dispense Checkout

Use Case ID: UC-01

Use Case Name: Process Sale / Dispense Checkout

Primary Actor: Pharmacy Clerk

Secondary Actor: Stock Monitor — a system process that checks the stock level after a sale.

Related Requirements:

- FR-001- FEFO-based batch selection
- FR-002- Stock threshold monitoring

Trigger:

A patient comes to the pharmacy with a prescription or requests an OTC medicine, and the Pharmacy Clerk starts the checkout process.

Preconditions:

1. The Pharmacy Clerk must be logged into the dispensing system.
2. The requested medicine must have at least one batch that has not expired.
3. If the medicine requires a prescription, the prescription must have been verified.

Postconditions:

1. The stock of the selected medicine batch is reduced by the quantity sold.
2. A sale record is created containing the clerk's ID, medicine SKU, batch ID, quantity sold, and time of sale.
3. The Stock Monitor is notified so that it can check whether the medicine has fallen below its safety stock level.

Main Success Scenario

1. The Pharmacy Clerk searches for the medicine requested by the patient.
2. The system shows the available batches of that medicine along with their current stock quantities.
3. The Clerk enters the quantity required and proceeds with the checkout.
4. The system applies **FEFO (First-Expired-First-Out)** and selects the available batch that expires soonest, as long as it has enough stock to fulfil the request.
5. The system reduces the stock of the selected batch by the quantity sold.
6. The system records the sale details and displays or prints the receipt.
7. The system informs the Stock Monitor that the sale has been completed so it can check the medicine's stock level.
8. The checkout is completed successfully.

Alternate Flow A1: Selected Batch Does Not Have Enough Stock

Trigger: This occurs at Step 4 when the batch with the nearest expiry date does not have enough stock to fulfil the entire request.

1. The system notices that the selected batch does not have enough medicine.
2. The system uses all the remaining stock from that batch and then selects the next batch that expires soonest for the remaining quantity.
3. The system shows the Clerk how the requested quantity will be taken from the different batches.
4. The Clerk reviews and confirms the allocation.
5. The system updates the stock of all the batches involved and creates one sale record containing the details of each batch used.
6. The process continues from Step 6 of the main scenario.

Exception Flow E1: No Valid Batch Available

Trigger: This occurs at Step 4 when there is no non-expired batch with enough combined stock to fulfil the request.

1. The system checks all available batches and finds that none can be used because they are either expired or do not have enough stock.
2. The system stops the checkout for that medicine and informs the Clerk that the medicine is unavailable.
3. The system records the failed dispensing attempt for auditing and alerts the Stock Monitor.
4. The checkout ends without creating a sale for that medicine.